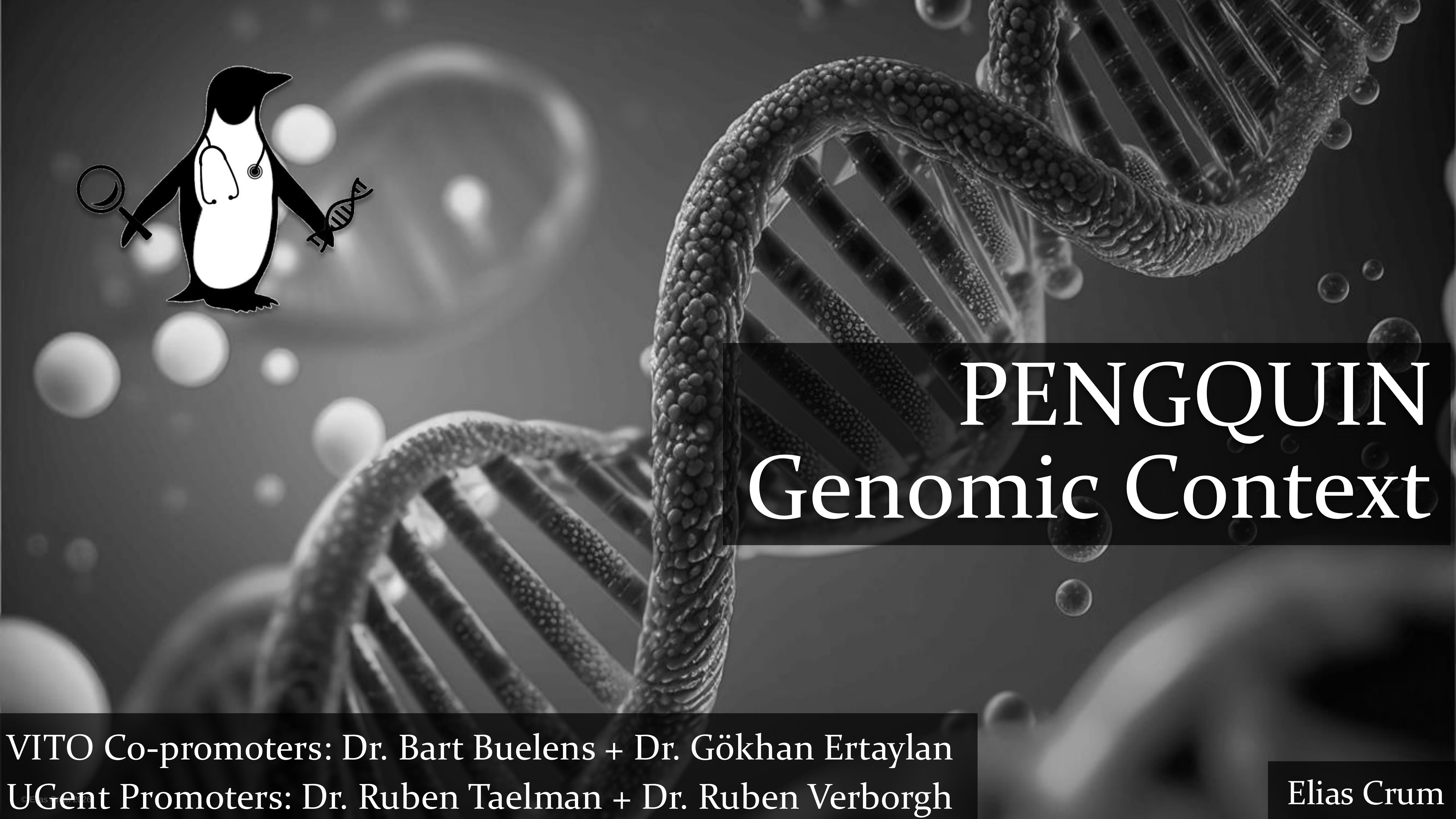

PENGQUIN
Genomic Context
VITO Co-promoters: Dr. Bart Buelens + Dr. Gökhan Ertaylan
UGent Promoters: Dr. Ruben Taelman + Dr. Ruben Verborgh
Elias Crum

Agenda

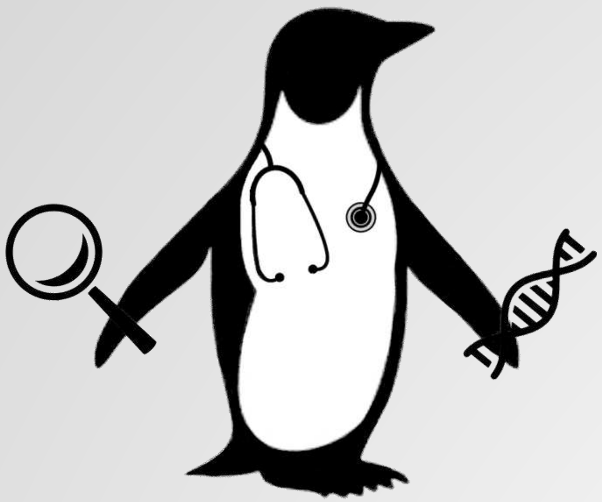
1. PENGQUIN Project Introduction
2. Introduction to Genome Sequencing
3. Genome Data
4. Current Genomic Data Storage and Querying Strategies
5. Genomic Data as Linked Data?

Things to note

1. If you have a question at <u>any</u> point, PLEASE ASK!!

2. I will ask “does this make sense” – I mean it (not just rhetorically)

3. This presentation is VERY HIGH LEVEL



PENGQUIN

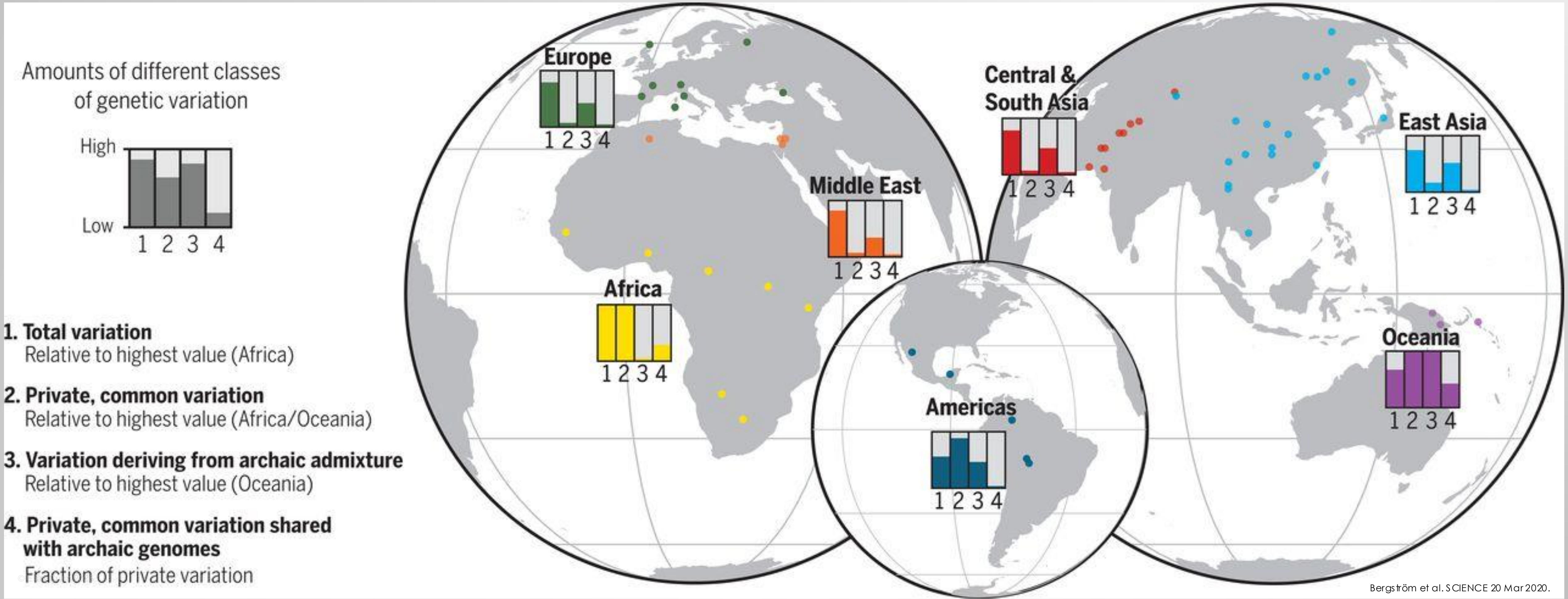
PErsoNal Genome QUery IN healthcare and clinical practice

Human Genome Variation

Global genetic variation (929 genomes)

- 67.3 million single-nucleotide polymorphisms
- 8.8 million small insertions or deletions (indels)
- 40,736 copy number variants

All humans are 99.9% genetically identical



Bergström et al. SCIENCE 20 Mar 2020.

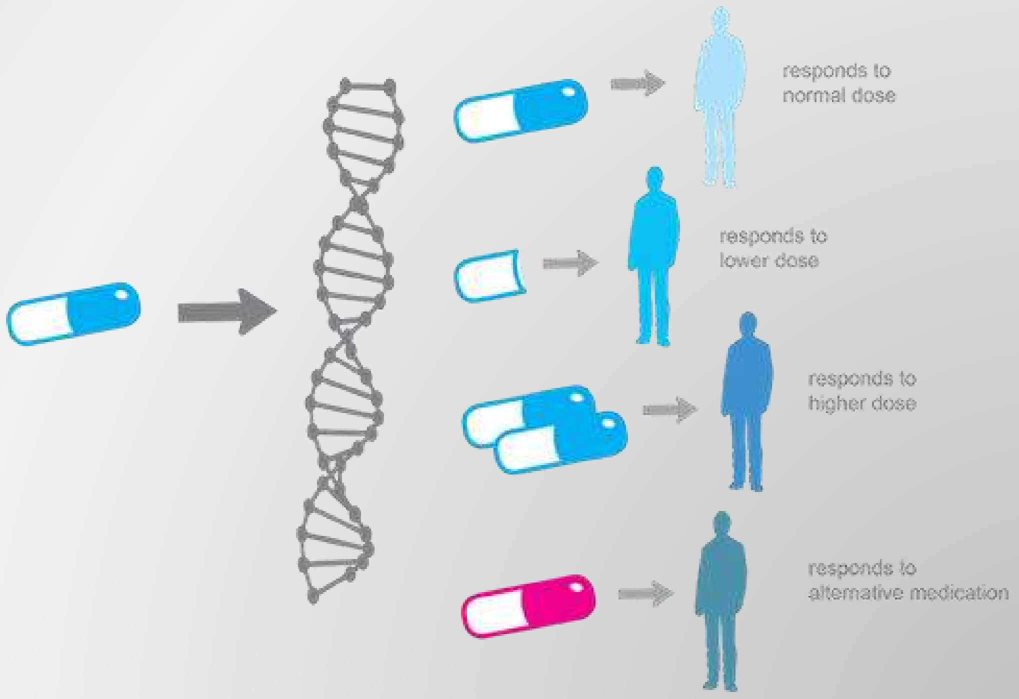
Personalized Medicine

All human diseases are influenced by genetics

Personalized Medicine = Medical care informed by an individual's genetic profile

Examples:

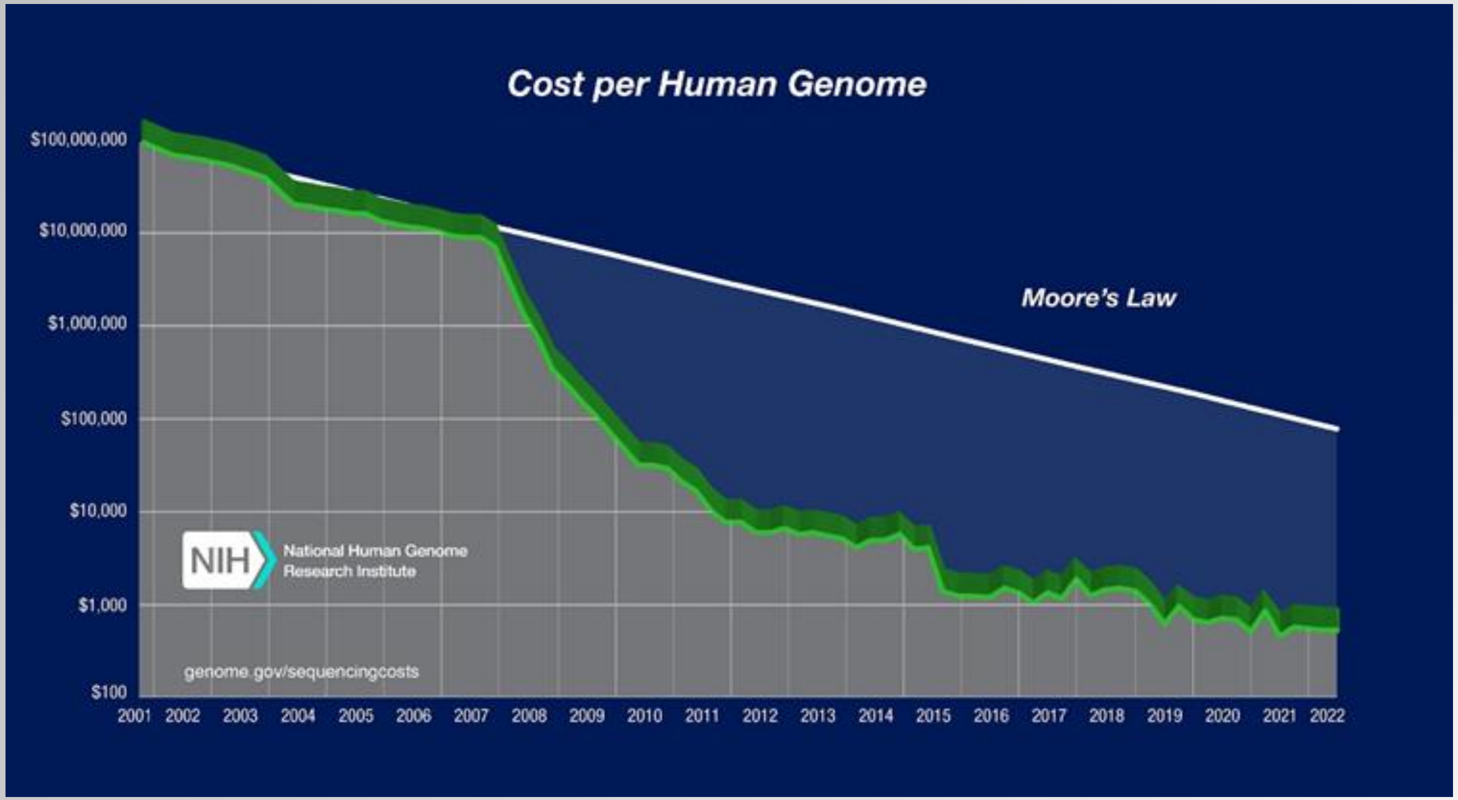
- Drug prescription
- Cancer diagnosis
- Cancer therapy development and targeting
- Genetic disease screening



Jackson et al. Essays Biochem.3 Dec 2018.

https://alphagenomix.com/empowering-personalized-medicine/

Sequencing Costs



https://www.genome.gov/about-genomics/fact-sheets/Sequencing-Human-Genome-cost

Currently ~€300 per genome

Per-base cost of **sequencing** has been dropping by half every ~5 months

Research Context

Challenges to personalized medicine?

↓

Genome storage and querying

↓

Why?

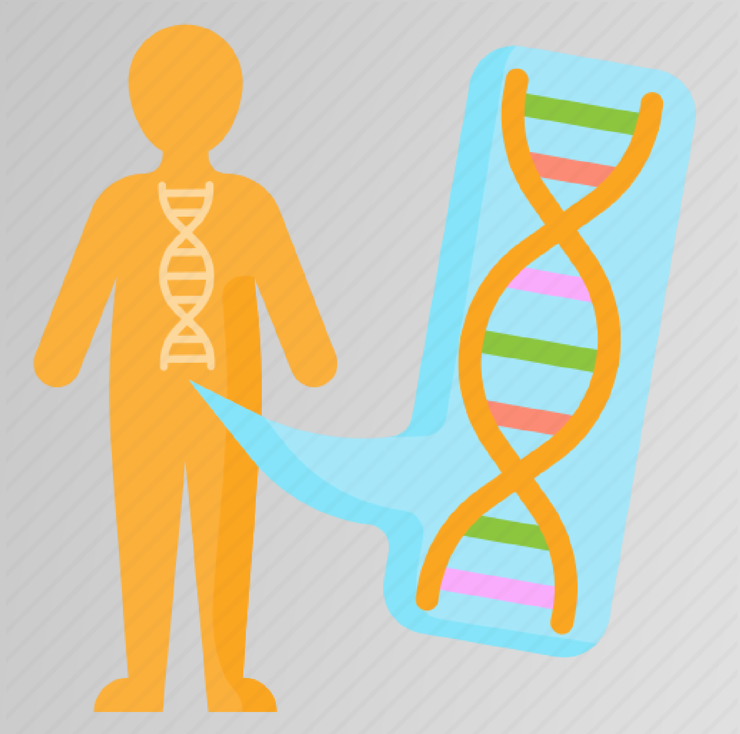
Privacy concerns

~3 BILLION
base pairs

Genome storage

Stored data querying

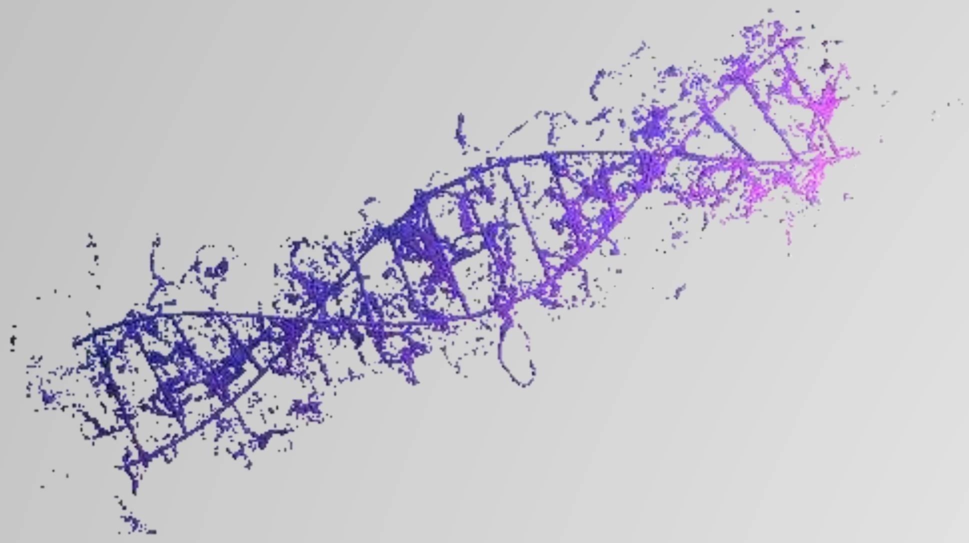
How do we go from:

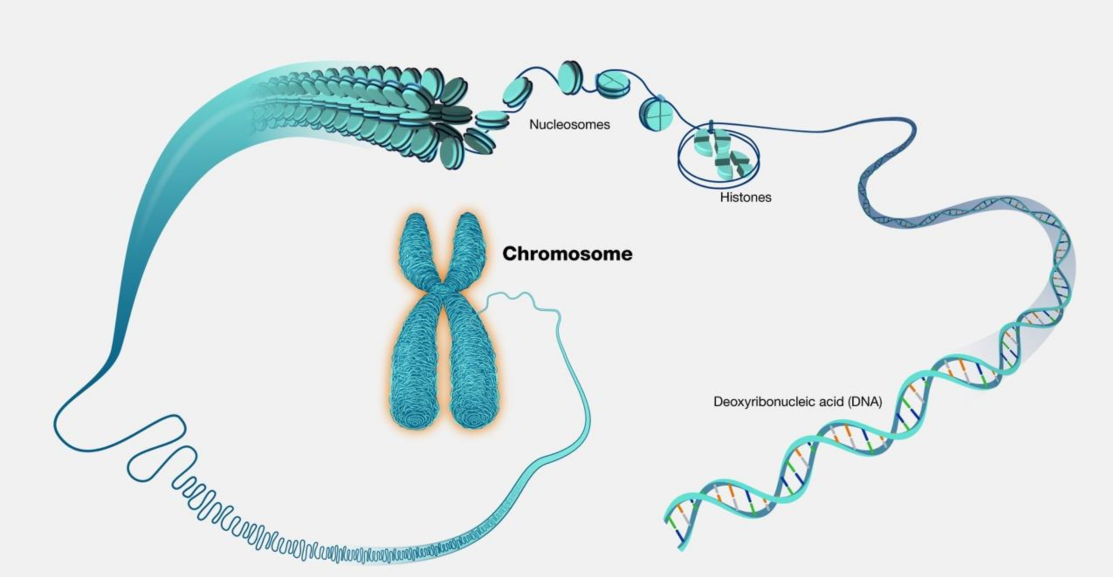


```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

To this?

The VERY basics:





Deoxyribonucleic Acid (DNA):

- 2 strands (that are complementary to each other)
- At each position contains a nucleotide (base)
  - A , T , C , G
  - A-T / C-G
- Wound tightly together into chromosomes
- Within EVERY living cell of your body (other than red blood cells)
- Each human has ~108 x 10^9 **kilometers** of DNA

https://www.genome.gov/genetics-glossary/Chromosome

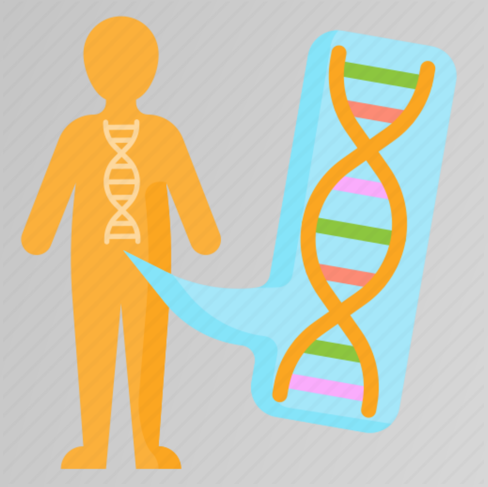
https://www.nigms.nih.gov/education/Inside-Life-Science/Pages/Genetics-by-the-Numbers.aspx

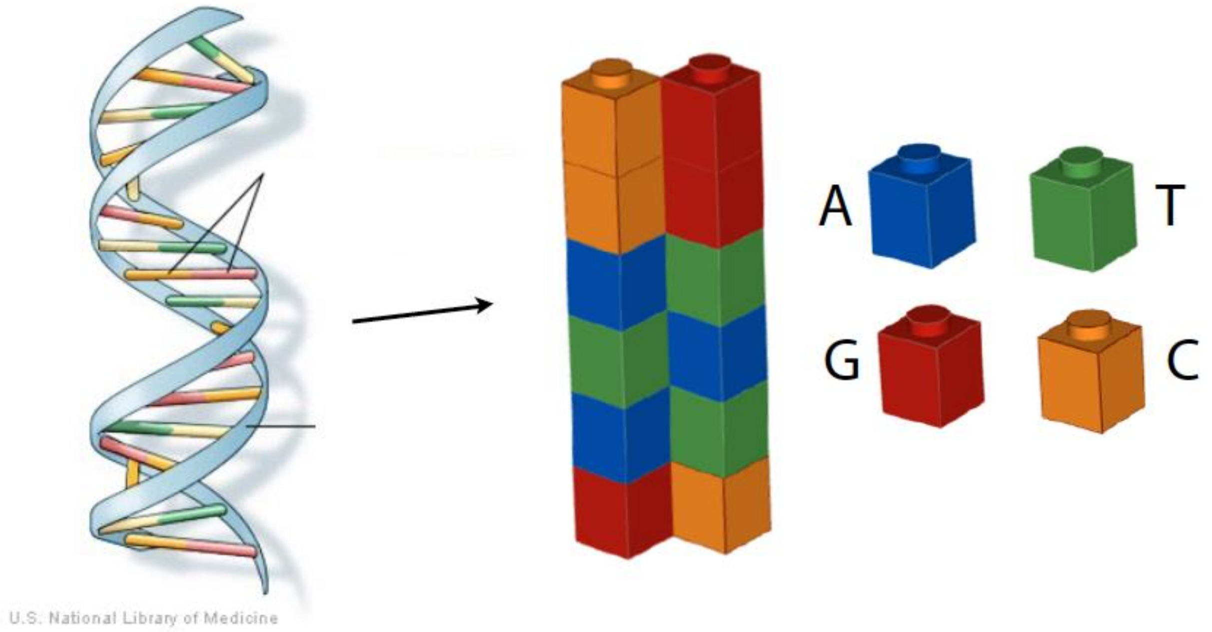
Genome Sequencing

Three Generations of Sequencing Technologies:

- 1st Gen: Sanger Sequencing (~2000s)
- 2nd Gen: Shotgun Sequencing (~2005 – Current)
- 3rd Gen: Long-read Sequencing (2015 – Current)

Quick Genome Sequencing Overview





Most cost-effective strategy currently:

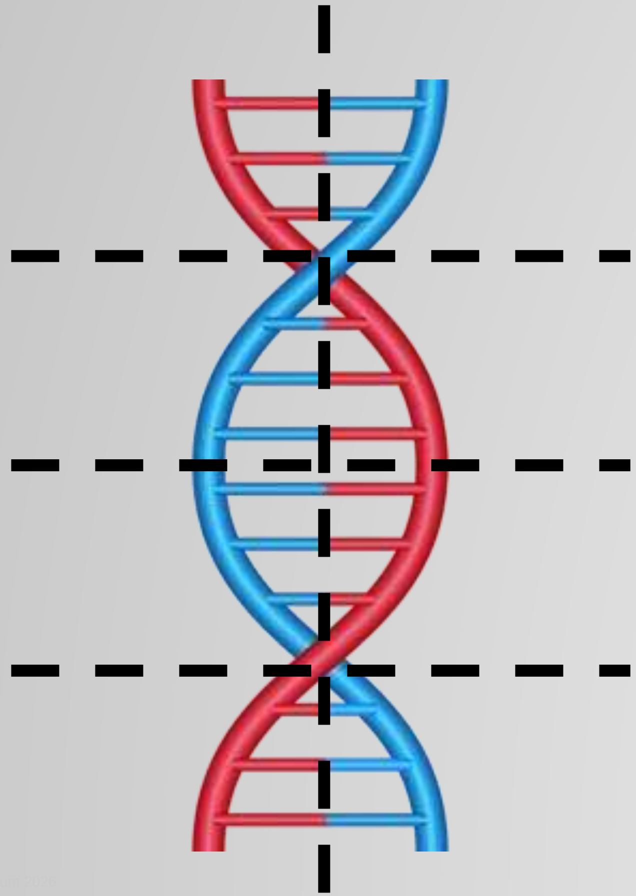
- 2nd Gen Sequencing
  - Short Reads (~150 base pairs)
  - High Throughput (>30x representation of each base in genome)

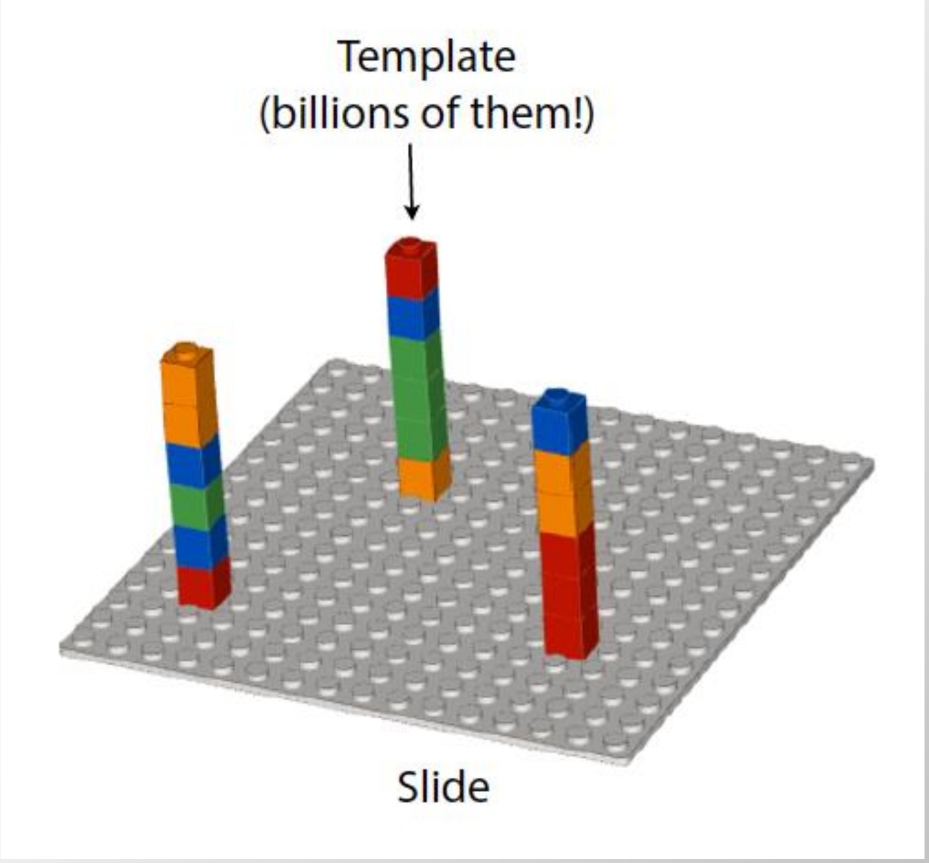
from Ben Langmead, PhD

2^{nd} Gen Sequencing Preprocessing

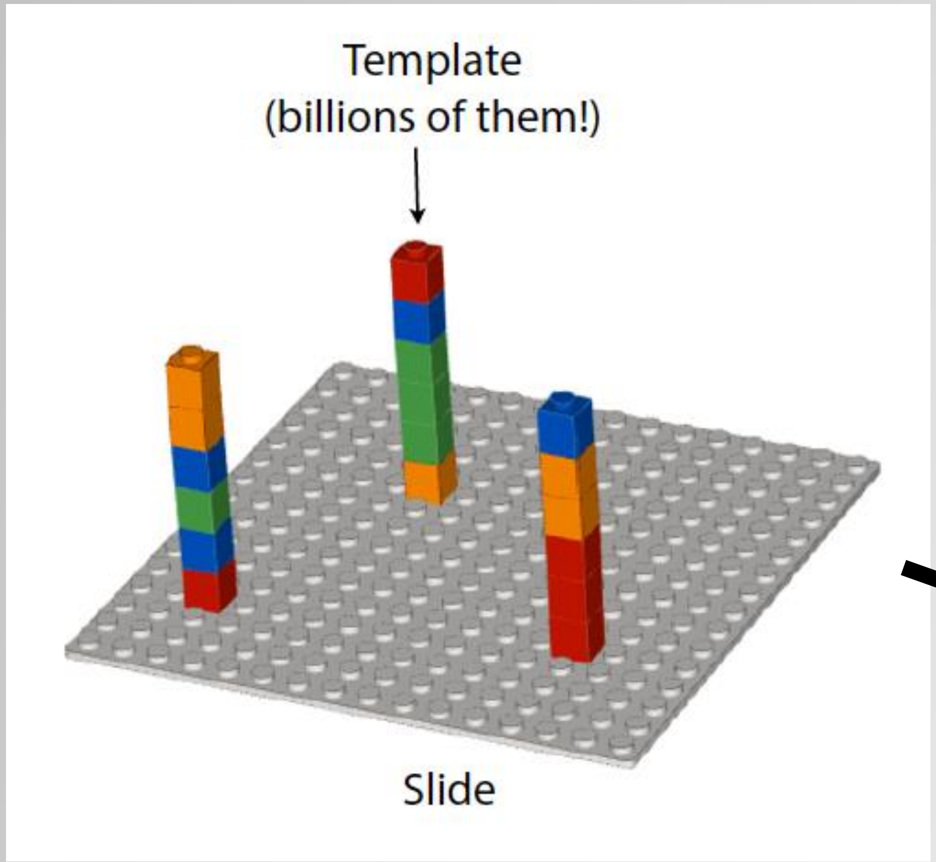
** Most cost-effective strategy currently:

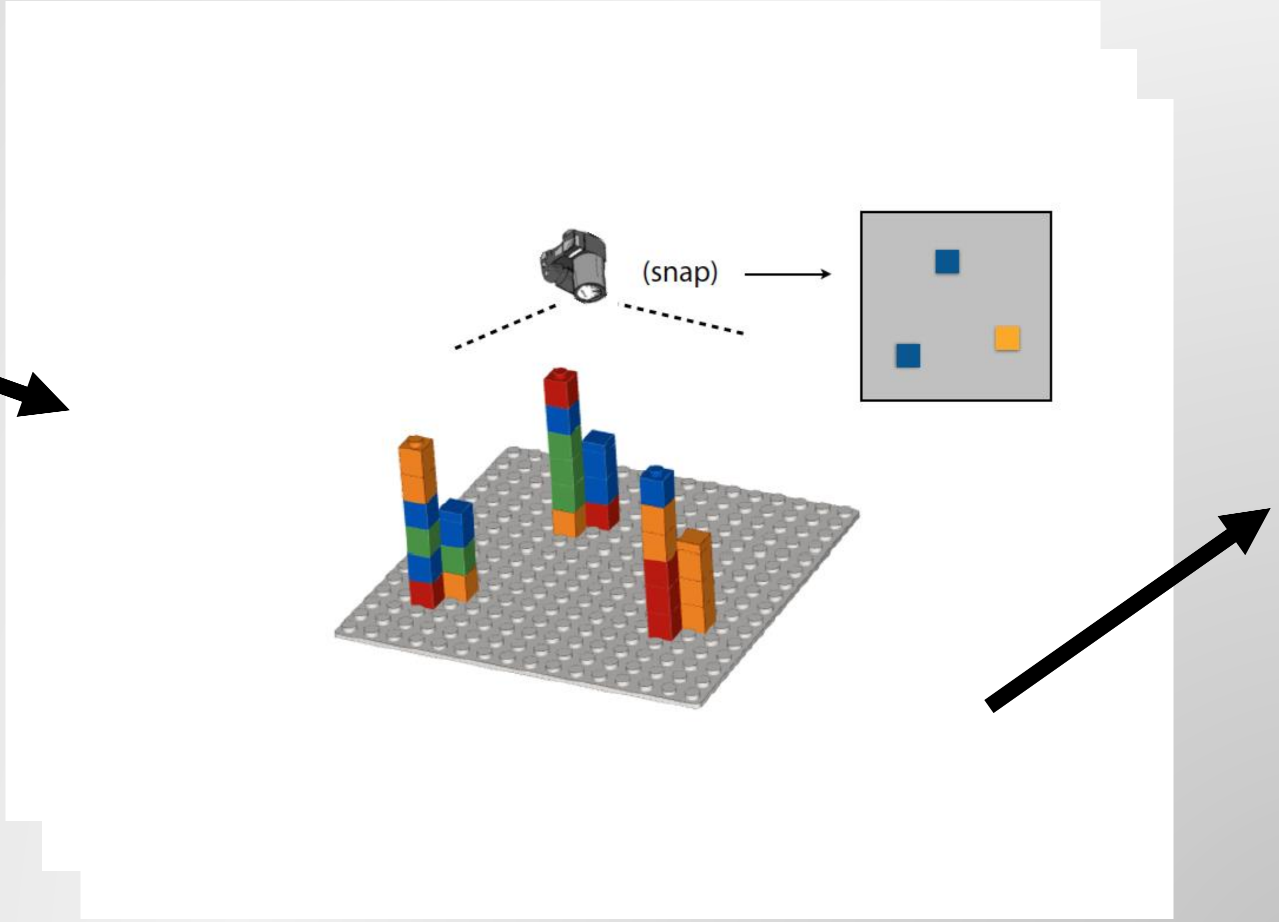
Short Reads (~100 - 150 base pairs)



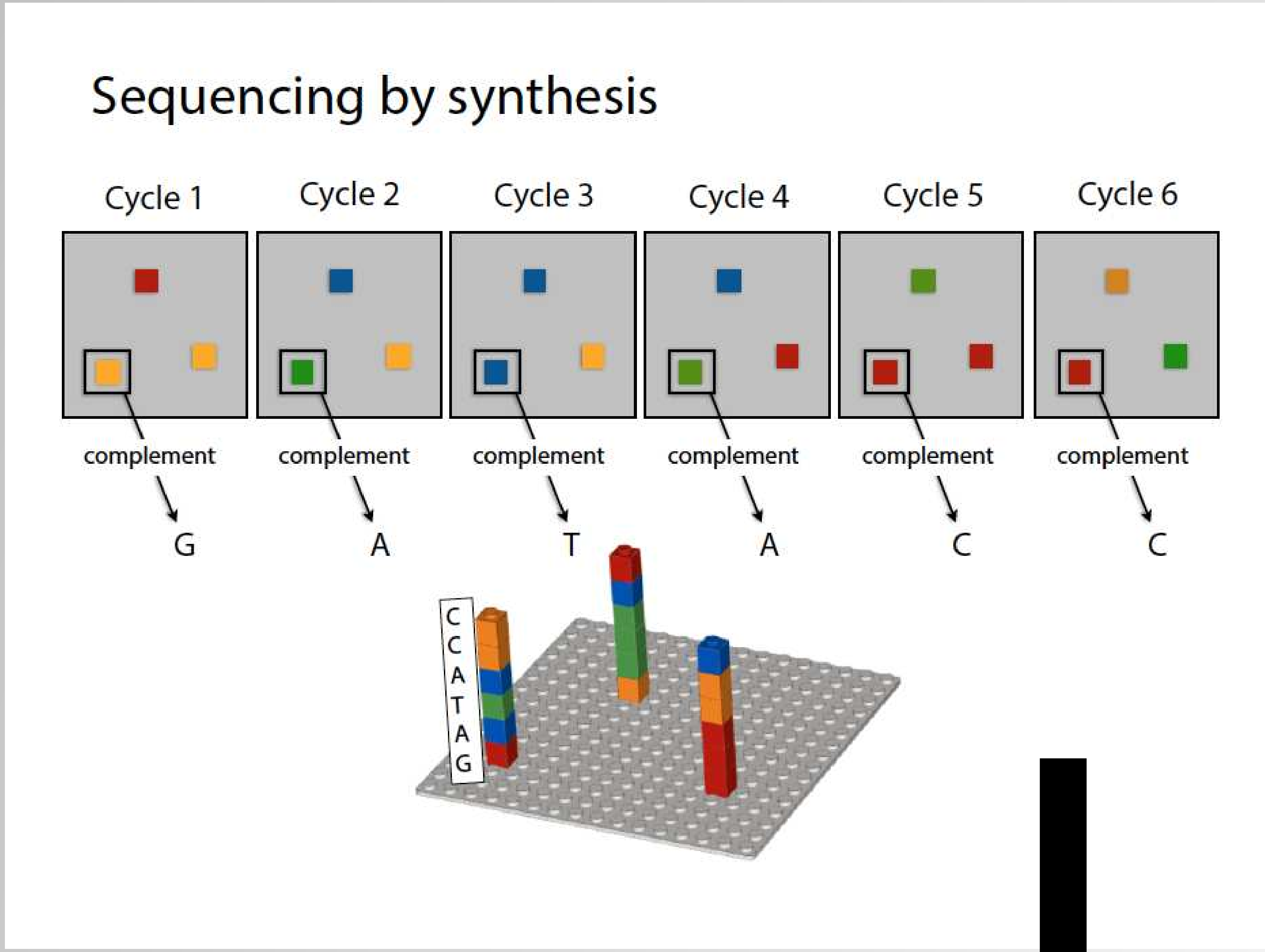


Quick Genome Sequencing Overview





Quick Genome Sequencing Overview



Video with slightly more detail: [here]

```
@K00282:309:HWT23BBXX:4:1101:1610:1086
TGGGAGTGTCTTCAGCCCTAAGGCAGCATCAGAGATCATAATCTTCTGCTC
+
AAAFFJ-AJFJFFJFJFFJ<JJAFJJFFJJJJAFA<AJFJFJFF<JJFJF<
@K00282:309:HWT23BBXX:4:1101:1671:1086
CCAGAAGGAGGCAGAGCTCCGCACCTGGATCGAGGGACTCACCGGCCTCTC
+
-<AFFJJAJF<AJJFJJJ<FJJJ7FJJJJ<JJJJJJFJFJJF<-FJJJJFJ
@K00282:309:HWT23BBXX:4:1101:2179:1086
GGGCACTCGGTACTGCTGCAGGAGCTGTTGCTGGAAGTGGAACCCAGGCGG
+
--AAFJJFAJA7A-F<-FJF<AAAJFF--F-7FJ-<FFFJ<-A777AFAFF
@K00282:309:HWT23BBXX:4:1101:2402:1086
ACCATCACCCTCGAGGTTGAACCCTCGGATACGATAGAAAATGTAAAGGCC
+
AAAFFJJJFJJFFJJJJJJJJJJJJJJJJJJJJJFFJJFJJ-FJJJJJFJJJ
@K00282:309:HWT23BBXX:4:1101:2625:1086
CGGAGACAGGGCAGACTGTGCCCATCTCCTCTCCTCACCAGCTGTGTACCT
+
<AF-F-FAFJJF<J<<A<AAJ7FJJJJJJJJJ<J-<<---AAAJJJJJFJ7
```

Sequence Data Formats

FASTQ ✚ **FASTA** → **SAM/BAM** → **VCF**

Raw Sequencing Outputs

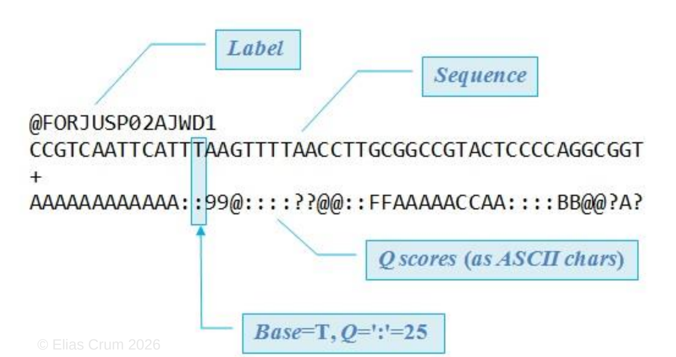
Contiguous DNA sequence

FASTQ reads aligned to a FASTA reference

Areas in SAM where sample and reference differ

```
>gi|568336023|gb|CM000663.2| Homo sapiens chromosome 1, GRCh38 reference primary assembly
CCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTA
ACCCTAACCCTAACCCTAACCCTAACCCAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCC
TAACCCTAACCCTAACCCTAACCCTAACCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACC
CTAACCCTAACCCCTAACCCTAACCCTAAACCCTAAACCCTAACCCTAACCCTAACCCTAACCCTAACCC
CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCTAACCCCTAACCCTAACCCTAACCCTACCCTAAC
CCTAACCCTAACCCTAACCCTAACCCTAACCCCTAACCCCTAACCCTAACCCTAACCCTAACCCTAACCC
TAACCCTAACCCCTAACCCTAACCCTAACCCTAACCCTCGCGGTACCCTCAGCCGGCCCGCCCGCCCGGG
TCTGACCTGAGGAGAACTGTGCTCCGCCTTCAGAGTACCACCGAAATCTGTGCAGAGGACAACGCAGCTC
CGCCCTCGCGGTGCTCTCCGGGTCTGTGCTGAGGAGAACGCAACTCCGCCGTTGCAAAGGCGCGCCGCGC
CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGC
AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGC
CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCGGGGTGGAGGCGTGGCGCAGGCGCAGAGAGGCGC
GCCGCGCCGGCGCAGGCGCAGAGACACATGCTACCGCGTCCAGGGGTGGAGGCGTGGCGCAGGCGCAGAG
AGGCGCACCGCGCCGGCGCAGGCGCAGAGACACATGCTAGCGCGTCCAGGGGTGGAGGCGTGGCGCAGGC
GCAGAGACGCAAGCCTACGGGCGGGGGTTGGGGGGGCGTGTGTTGCAGGAGCAAAGTCGCACGGCGCCGG
GCTGGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAAACTCACGTCACGGTGGCGCGGCGCAGAGACG
GGTAGAACCTCAGTAATCCGAAAAGCCGGGATCGACCGCCCCTTGCTTGCAGCCGGGCACTACAGGACCC
GCTTGCTCACGGTGCTGTGCCAGGGCGCCCCCTGCTGGCGACTAGGGCAACTGCAGGGCTCTCTTGCTTA
GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGGCACTGCAGGGCCCTCTTGCTTACTGTATAGTGGTGG
```

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257 G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .         T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355 A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .         T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1 GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```




```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

Sequencing Storage

The human genome = ~3 billion base pairs

- 3 billion letters -- **~700 megabytes**

Representative of a **single strand** of the whole human genome sequence
E.g. : …ATGCCGTAAACAGATGTCA …

Format: *TXT* file

- Raw human whole genome sequence -- **~200 gigabytes**

Whole genome sequences typically have **>=30x coverage at each base position**. (i.e. each base of the genome is represented in the sequence file a >= 30 times)

Format: *FASTQ* file

- Aligned human whole genome sequence -- **~210 gigabytes // ~100 gigabytes // ~15 gigabytes**

Whole genome sequence **is mapped to a reference human genome** and can be stored in various formats. Recent advances have allowed for improved compression.

Format*: SAM* file // *BAM* file // *CRAM* file

- Human genome variant file (list of mutations) -- **~125 megabytes**

Individual humans only exhibit **~0.1% variation** comparatively (~3 million mutations). Variant file **records those differences** in one individual’s genome compared to the “normal” reference genome.

Format: *VCF* file

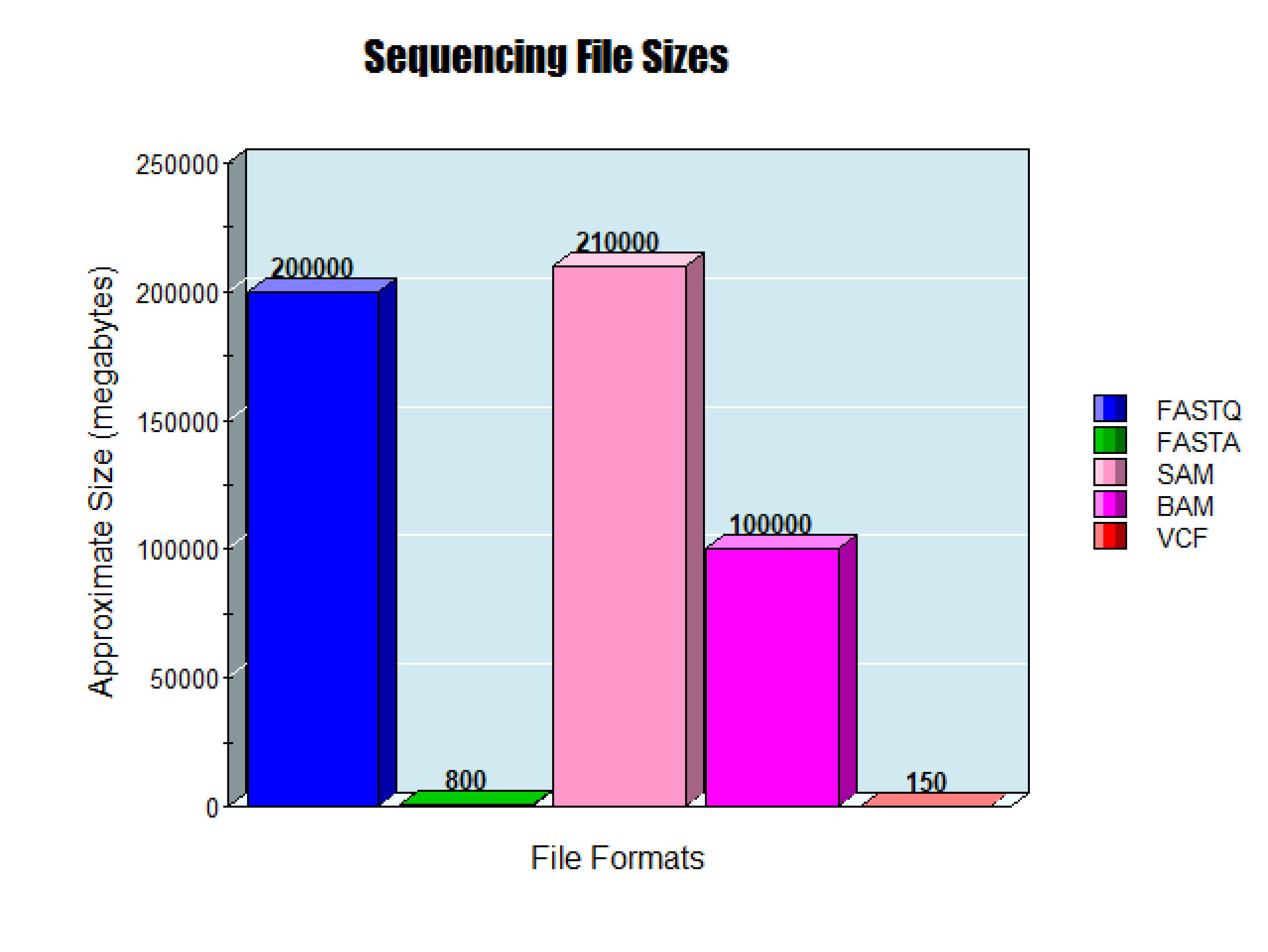
Sequencing File Sizes

FASTQ ~ 200 **giga**bytes

FASTA ~ 800 **mega**bytes

SAM/**BAM** ~ 210/100 **giga**bytes

VCF ~ 150 **mega**bytes



https://nces.ed.gov

Variant Call Format (VCF)

- tab delimited .txt file

- Rows represent individual mutations in a sequence (when compared to reference)

- Columns store information about that mutation

```
#CHROM  POS  ID  REF  ALT  QUAL  FILTER  INFO  FORMAT
```

```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file://23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM  POS     ID          REF ALT QUAL    FILTER  INFO    FORMAT  GEN
chr1    82154   rs4477212   a   .   .   .   .   GT  0/0
chr1    752566  rs3094315   g   A   .   .   .   GT  1/1
chr1    752721  rs3131972   A   G   .   .   .   GT  1/1
chr1    798959  rs11240777  g   .   .   .   .   GT  0/0
chr1    800007  rs6681049   T   C   .   .   .   GT  1/1
chr1    838555  rs4970383   c   .   .   .   .   GT  0/0
chr1    846808  rs4475691   C   .   .   .   .   GT  0/0
chr1    854250  rs7537756   A   .   .   .   .   GT  0/0
chr1    861808  rs13302982  A   G   .   .   .   GT  1/1
chr1    873558  rs1110052   G   T   .   .   .   GT  1/1
chr1    882033  rs2272756   G   A   .   .   .   GT  0/1
chr1    888659  rs3748597   T   C   .   .   .   GT  1/1
chr1    891945  rs13303106  A   G   .   .   .   GT  0/1
```

Variant Call Format (VCF)

* Important note: <u>1</u> VCF file <u>CAN</u> represent mutations of <u>MANY</u> sequences

```
##fileformat=VCFv4.2
##fileDate=20151002
##source=callMomV0.2
##reference=gi|251831106|ref|NC_012920.1| Homo sapiens mitochondrion, complete genome
##contig=<ID=MT,length=16569,assembly=b37>
##INFO=<ID=VT,Number=.,Type=String,Description="Alternate allele type. S=SNP, M=MNP, I=Indel">
##INFO=<ID=AC,Number=.,Type=Integer,Description="Alternate allele counts, comma delimited when multiple">
##FILTER=<ID=fa,Description="Genotypes called from fasta file">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM  POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  HG00096 HG00097 HG00099 HG00100 HG00101 HG00102 HG00103 HG00105 HG00106 HG00107
MT  10  .   T   C   100 fa  VT=S;AC=3   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  16  .   A   T   100 fa  VT=S;AC=3   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  26  .   C   T   100 fa  VT=S;AC=3   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  35  .   G   A   100 fa  VT=S;AC=2   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  40  .   TC  CT  100 fa  VT=M;AC=1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  41  .   C   T   100 fa  VT=S;AC=4   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  42  .   TCC CCC,T   100 fa  VT=S,I;AC=1,1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  46  .   T   C   100 fa  VT=S;AC=1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  47  .   G   A   100 fa  VT=S;AC=1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  52  .   TGG CAA 100 fa  VT=M;AC=1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  55  .   TATTTT  T,CATTTT,AATTTT,TTT,TTTTT   100 fa  VT=I,S,S,I,I;AC=5,3,2,1,1   GT  0   0   0   0   0   0   0   0   0   0   0   0
MT  57  .   T   C   100 fa  VT=S;AC=3   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  58  .   TTT T   100 fa  VT=I;AC=4   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
MT  59  .   T   A   100 fa  VT=S;AC=1   GT  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
```

Existing VCF Querying Tools?

- VCFio
- Seqminer2
- BCFtools

Tabix --> hybrid of binning and linear index

Made to query 1.000+ VCF (and BCF) files all stored in the same location

Small VCF analysis can mostly be performed by brute-force parsing

Existing VCF Querying Tools?

Tabix Strategy:

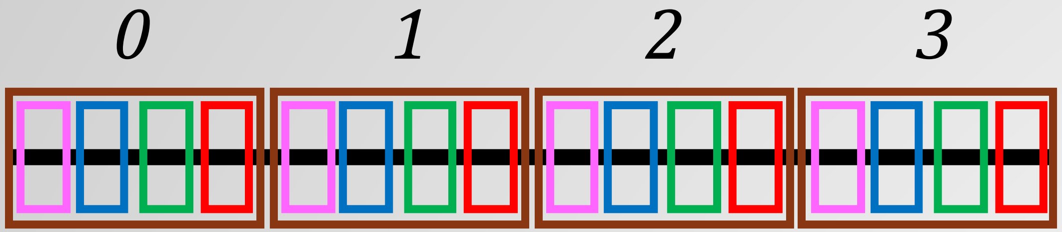
VCF File → Indexing → Index file

Shows where variants are stored (binary)

Mutations are placed into “bins”

A “bin” : Represents a subset of byte locations within a VCF file == Range of base positions on a chromosome

VCF File:



2048 bins at each level

1. Bins have a hierarchy

2. Bins contain a subset of mutations represented in the VCF file

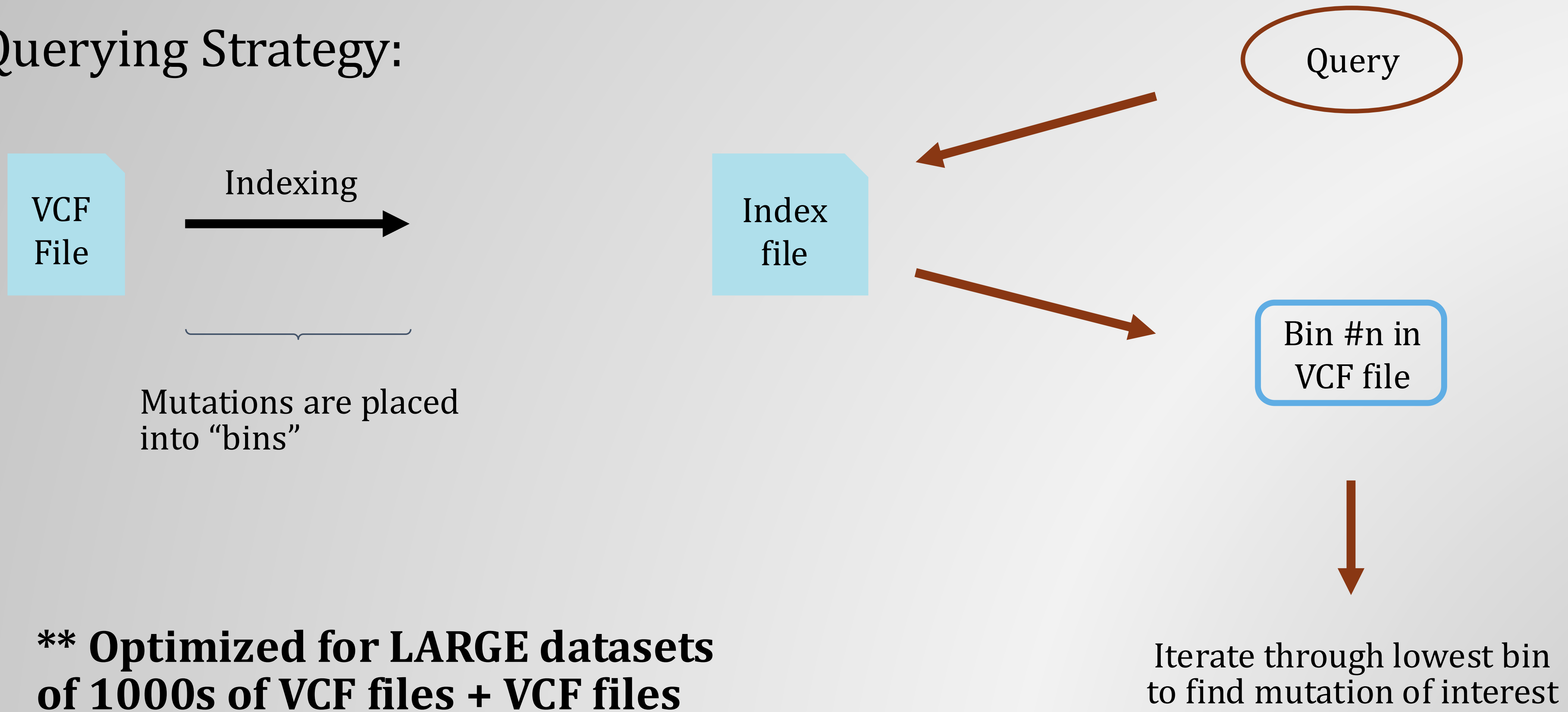
3. Bin memory size is determined by file size

4. Index contains information about mutation byte offset
   → specific look-up coordinates

5. Typically, VCF files are .gz compressed

Existing VCF Querying Tools?

Querying Strategy:



**** Optimized for LARGE datasets of 1000s of VCF files + VCF files representing MANY sequences ****

What if we used linked data strategies?

Architecture of data storage?

What will the links be?

How can we leverage the different architecture?

Query engines? And how will data transfer be handled?

Previous Implementation

VCF
File

Download + decompress
vcf.gz file

**** Serialization + Querying takes ~30 minutes ****

For more detail see https://github.com/ecrum19/vcf_processing_analysis/tree/main

Chr1.ttl Chr2.ttl Chr3.ttl …

Serialize VCF file into
RDF (ttl)

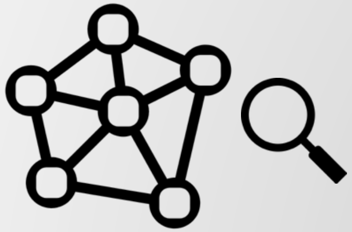
Query:

SELECT ?o

WHERE {

id:MutantID info:Variant ?o .

}



SPARQL query using
RDFLib

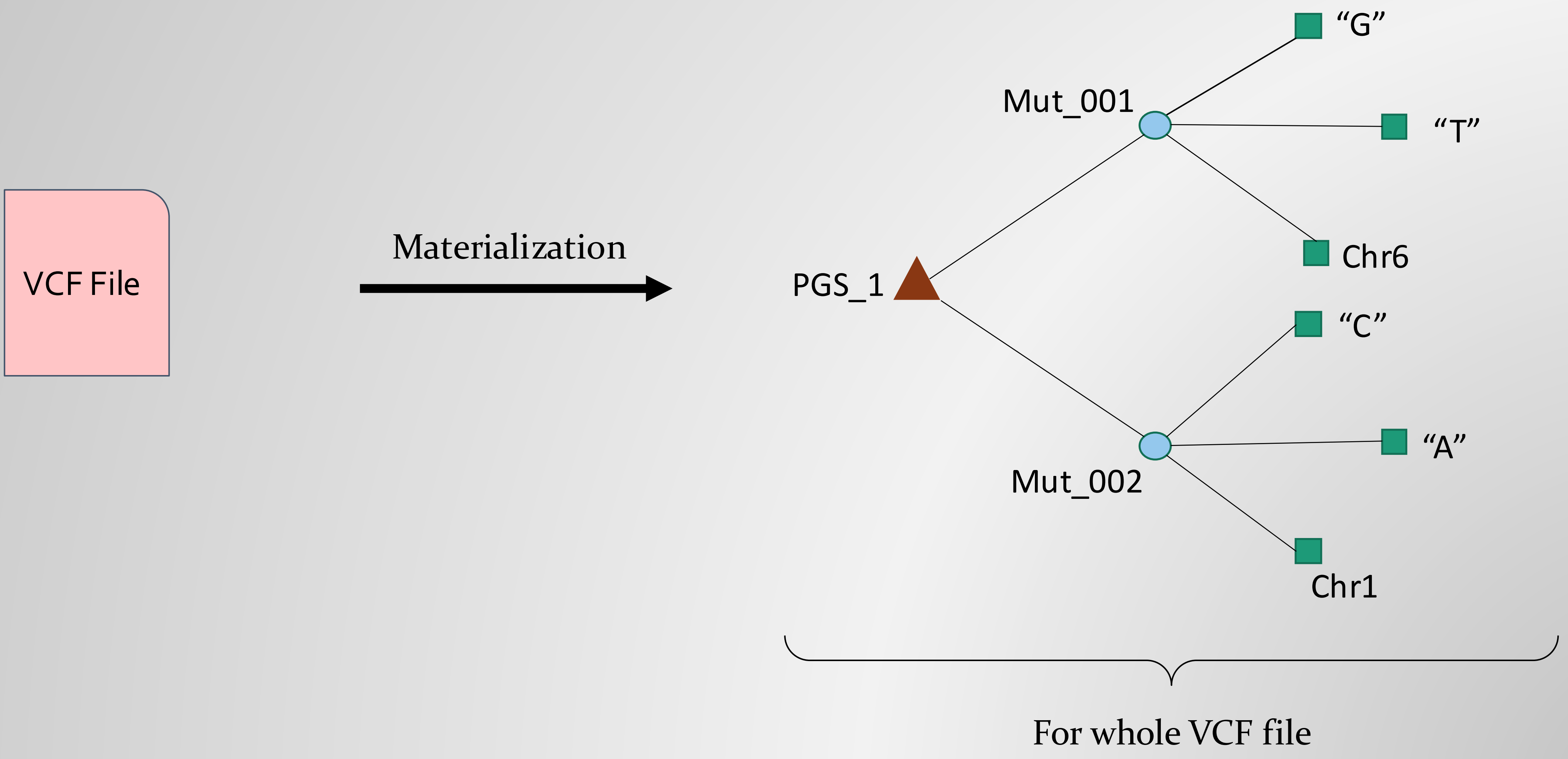
Potential Approach α.

= literal

= internationalized resource identifier (IRI)

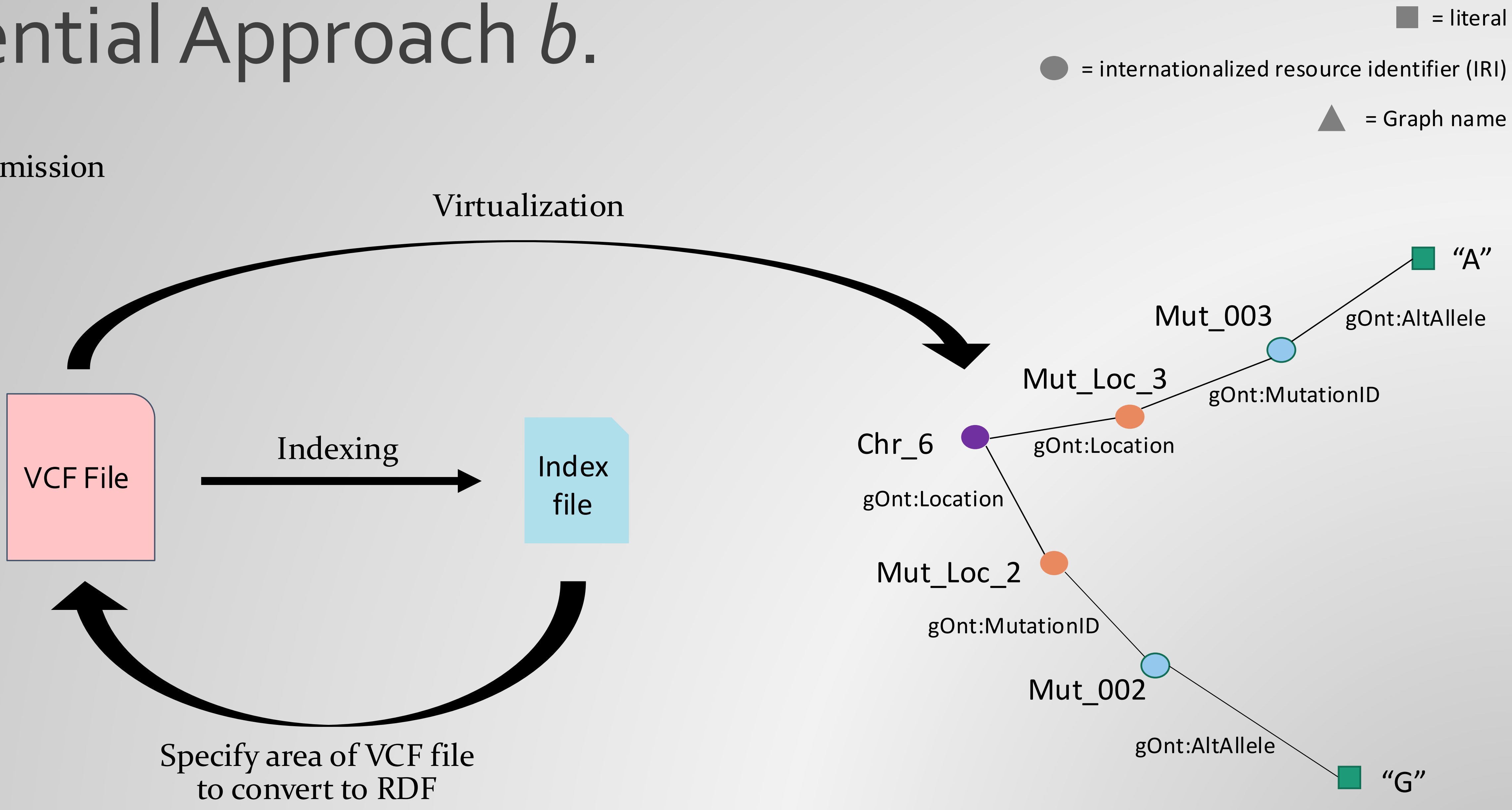
Mut = mutation

* With permission



Potential Approach *b*.

* With permission



Questions?